MailFlow

Frontend Specification Document (FSD)

1. Purpose

This document defines the frontend architecture, UI structure, reusable components, design principles, navigation, state management, and user experience guidelines for the MailFlow platform.

2. Frontend Technology Stack

React.js with Vite, Tailwind CSS, React Router, Axios, React Hook Form, Zustand (or Context API for MVP), React Query (future), and Framer Motion for subtle animations.

3. Design Principles

Maintain a clean, modern SaaS interface inspired by platforms such as Attio and Linear. Focus on simplicity, accessibility, consistency, responsive layouts, and minimal visual clutter.

4. Application Layout

The application consists of a persistent left sidebar, top navigation bar, statistics cards, data tables, right-side drawers for editing and AI previews, and responsive layouts for desktop and mobile.

5. Navigation Structure

Dashboard, Leads, Campaigns, Templates, WhatsApp, Analytics, Settings, Profile and Authentication screens.

6. Component Architecture

Reusable UI components include Buttons, Inputs, Tables, Badges, Modals, Drawers, Cards, Statistics Widgets, Search Bars, Pagination, Toast Notifications, Loaders and Empty States.

7. Pages

Landing Page, Login, Register, Dashboard, Leads Management, Upload Leads, AI Research, Template Builder, Campaign Preview, Analytics Dashboard, Settings and Profile.

8. State Management

Global state manages authentication, user profile, campaigns and theme. Local state manages forms, filters, drawers and temporary UI interactions.

9. API Integration

Frontend communicates with REST APIs using Axios. Authentication tokens are attached automatically and errors are handled globally.

10. Responsive Design

Desktop-first design with responsive layouts for tablets and mobile devices. Tables collapse gracefully and drawers become full-screen on smaller screens.

11. Performance

Code splitting, lazy loading, image optimisation, caching, memoisation and efficient rendering are used to improve performance.

12. Accessibility

Semantic HTML, keyboard navigation, sufficient contrast ratios, focus indicators and screen reader compatibility should be followed.

13. Future Enhancements

Dark mode customisation, internationalisation (i18n), drag-and-drop campaign builder, keyboard shortcuts and offline support.

Frontend Technology Summary

Layer	Technology
Framework	React.js + Vite
Styling	Tailwind CSS
Routing	React Router
State Management	Zustand / Context API
Forms	React Hook Form
HTTP Client	Axios
Animation	Framer Motion
Icons	Lucide React
Notifications	React Hot Toast